Causal Inference in Epidemiology

Douglas Ezra Morrison

Last modified: 2026-11-10: 0:27:36 (UTC)

Contents

Preface 1
- UC Davis courses . . . 1
- Online Videos . . . 1
- UC Davis Datalab learning group . . . 2
- Books . . . 2

1 Introduction to causal inference 3
- 1.1 Introduction . . . 3
- 1.2 Individual causal effects . . . 3

2 Philosophy of causality 6
- 2.1 Why look at philosophy at all? . . . 6
- 2.2 Five accounts of causation . . . 6
- 2.3 What the potential-outcomes framework takes from these accounts . . . 8
- 2.4 Not everyone agrees . . . 9
- 2.5 Further reading . . . 9

3 Collider bias 10
- 3.1 Definition . . . 10
- 3.2 Classic example: Sackett's hospitalized patients . . . 10
- 3.3 Modern example: The obesity paradox . . . 11
- 3.4 Prevention . . . 11

4 Bias analyses 12

5 Difference in differences analyses 13
- 5.1 Change in Changes . . . 13

6 Further resources 16
- 6.1 The Catalog of Bias . . . 16

7 Summary 17

References 18

Preface

This will be an online book about causal inference.

Here are some other resources for learning causal inference:

UC Davis courses

- EPI 205[1] "Principles of Epidemiology"
- EPI 206[2] "Epidemiologic Study Design"
- EPI/SPH 207[3] "Advanced Epidemiologic Methodology"
- EPI 225[4] "Advanced Topics in Epidemiology Methods"
- POL 285[5] "Statistics of Causal Inference in Political Science"
- MGB/MGP/MGT 454A[6] "Causal Inference and Statistical Experiments"
  - syllabus: https://webapps.aws.ucdavis.edu/public/documents/4861649/Syllabushttps://schedule.aws.ucdavis.edu/public/documents/5319910/Syllabus
- PSC 204B[7] "Causal Modeling of Correlational Data"
- PSC 205C[8] "Structural Equation Modeling"

Course search options:

- https://schedule.aws.ucdavis.edu/courseScheduling
- https://catalog.ucdavis.edu/course-search/
- https://catalog.ucdavis.edu/courses-subject-code/

Online Videos

- "Introduction to Causal Inference"[9] (slides here[10])
- Online Causal Inference Seminar series[11]

[1] https://catalog.ucdavis.edu/search/?q=EPI+205
[2] https://catalog.ucdavis.edu/search/?q=EPI+206
[3] https://catalog.ucdavis.edu/search/?q=EPI+207
[4] https://catalog.ucdavis.edu/search/?q=EPI+225
[5] https://catalog.ucdavis.edu/search/?q=POL+285
[6] https://catalog.ucdavis.edu/search/?q=MGB+454A
[7] https://catalog.ucdavis.edu/search/?q=PSC+204B
[8] https://catalog.ucdavis.edu/search/?q=PSC+205C
[9] https://ucdhs.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=43e9eb6f-3ed9-41ac-8ad9-ae22016572c8%20
[10] https://health.ucdavis.edu/media-resources/ctsc/documents/pdfs/causal-inference-intro-2022.pdf
[11] https://www.youtube.com/channel/UCiiOj5GSES6uw21kfXnxj3A/videos

UC Davis Datalab learning group

- https://datalab.ucdavis.edu/causal-inference/
  - Reading list[12]

Other links:

- https://cameron.econ.ucdavis.edu/causal/
- https://datalab-icmat.github.io/causal_reading_group.html
- Lab exercises by Ben Noble[13]: https://github.com/bennoble/causal-inference-2022

Books

- Judea Pearl (2016)
- Hernán and Robins (2020)

[12] https://docs.google.com/document/d/1K0QZFSjQIYnOTahpRK7Q83eaiIIFjfa-clSbj_ifgco/edit?tab=t.0#heading=h.farbmh6n76gq
[13] https://benjaminnoble.org/

1 Introduction to causal inference

1.1 Introduction

Hernán and Robins (2020) (page vii):

> Unfortunately, the scientific literature is plagued by studies in which the causal question is not explicitly stated and the investigators' unverifiable assumptions are not declared. This casual attitude towards causal inference has led to a great deal of confusion.

Rigorously defining cause and effect is difficult. Fortunately, many humans have strong intuitions about these concepts. We will make cursory attempts at definitions for the basic terms, and leave the finer points to philosophers.

1.2 Individual causal effects

Definition 1.1 (Action, intervention, exposure, policy, treatment)**.** An **action** (also called an **intervention, exposure, policy,** or **treatment**) is a choice that we consider making.

Definition 1.2 (Random variable)**.** A **random variable** is a variable that may have different values for different individuals and/or for different actions/exposures that an individual might experience. [1]

Definition 1.3 (potential outcome, consequence)**.** A **potential outcome** of action a (also called a **consequence** of a) is the value of a random variable Y that would occur if we were to take action a. The potential outcome of action a on random variable Y is often denoted $Y(a)$, Y^a, or $Y^{A=a}$. We will use notation $Y(a)$.

---

[1] adapted from Hernán and Robins (2020) p3

Definition 1.4 (Factual outcome, observed outcome)**.** A *factual outcome* (or *observed outcome*) is the potential outcome corresponding to an action that was actually taken.

If we consider taking action a or an alternative a', and we actually take action $A = a$, then $Y(a)$ is the factual outcome.

Definition 1.5 (Counterfactual outcome)**.** A counterfactual outcome is a potential outcome corresponding to an action that was not actually taken.

If we consider taking action a or an alternative a', and we actually take action $A = a$, then $Y(a')$ is a counterfactual outcome.

There might be more than one counterfactual outcome, depending on how many action options were considered, but there can only ever be one factual outcome per random variable.

Definition 1.6 (Cause)**.** Action a **causes** outcome y (or is a cause of y) if:

- outcome y would occur if we were to take action a
  and
- outcome y would not occur if we did not take action a.

In other words, if:

- $Y(a) = y$
  and
- $\exists a' \neq a : Y(a') \neq y$

Definition 1.7 (effect)**.** The effect of action a on outcome Y, relative to a given alternative action a', is the contrast in potential outcomes, $Y(a)$ versus $Y(a')$.

Definition 1.8 (Consistency)**.** Consistency is the assumption that if we observe an action a, then the observed outcome Y is equal to the "factual potential outcome" corresponding to action a; in other words, if $A = a$, then $Y(a) = Y$.

Definition 1.9 (Exchangeability)**.** Subpopulations defined by exposure X are exchangeable with respect to a potential outcome $Y(x)$ if the distribution of $Y(x)$ does not depend on the observed exposure X:

$$Y(x) \perp\!\!\!\perp X$$

Theorem 1.1. *If subpopulations defined by values of exposure X are exchangeable with respect to potential outcome $Y(x)$, then the expected value of $Y(x)$ does not depend on the observed value of X:*

$$E[Y(x)|X = x'] = E[Y(x)|X = x]$$

Definition 1.10 (Conditional exchangeability)**.** Subpopulations defined by exposure X are exchangeable with respect to a potential outcome $Y(x)$ if the distribution of $Y(x)$ does not depend on the observed exposure X, conditional on covariate(s) Z:

$$Y(x) \perp\!\!\!\perp X|\tilde{Z}$$

Theorem 1.2. *If subpopulations defined by values of exposure X are conditionally exchangeable with respect to potential outcome $Y(x)$ given covariate $\tilde{Z}$, then the expected value of $Y(x)$ does not depend on the observed value of X:*

$$E\Big[Y(x)|X = x', \tilde{Z} = \tilde{z}\Big] = E\Big[Y(x)|X = x, \tilde{Z} = \tilde{z}\Big]$$

Definition 1.11 (Fundamental problem of causal inference)**.** The *fundamental problem of causal inference* is that only one potential outcome (the factual outcome) can be observed per person (or per sampling unit, more generally) (Holland 1986). The other, counterfactual outcomes, are all missing data, and thus, the causal effects are all missing data as well.

2 Philosophy of causality

> **i** Optional reading
>
> This chapter is background. Nothing later in the book depends on it, but it explains where the potential-outcomes definitions in Chapter 1 come from, and why they carry the assumptions they do.

2.1 Why look at philosophy at all?

The book defines a cause in terms of potential outcomes (Definition 1.6). That definition is one answer among several to a question philosophers have argued over for centuries: what does it mean to say that one thing causes another? Seeing the other answers makes it easier to see what the potential-outcomes answer includes, what it leaves out, and why.

Each account below is a family of views, and this chapter simplifies all of them. The cited sources give the full arguments.

2.2 Five accounts of causation

2.2.1 Regularity

Definition 2.1 (Regularity view of causation)**.** On the **regularity view**, saying that A causes B means that events like A are always followed by events like B, with A coming first and the two occurring close together in space and time.

Hume's discussion of causation is the standard starting point (Hume 1748, sec. 7). His point is that we never observe a necessary connection between cause and effect. We observe only that one kind of event reliably follows another.

Example 2.1 (Regularity and smoking)**.** Under a strict reading, "smoking causes lung cancer" would be false, because many smokers never develop lung cancer and some non-smokers do.

Example 2.1 shows the main weakness: strict regularities are rare in epidemiology, where outcomes are stochastic and have many contributing causes. Mackie's refinement treats a cause as an insufficient but necessary part of a condition that is itself unnecessary but sufficient for the effect (Mackie 1965). That version fits "component causes" better.

2.2.2 Counterfactual dependence

Definition 2.2 (Counterfactual account of causation)**.** On the **counterfactual account**, event A causes event B if A and B both occurred and, had A not occurred, B would not have occurred.

Hume also wrote a second, counterfactual formulation of his definition (Hume 1748, sec. 7). Lewis later developed it into a full theory based on comparing the actual world with the closest possible worlds in which A does not occur (Lewis 1973).

This is the account closest to Definition 1.6. A potential outcome $Y(a')$ for an action a' that did not occur is a counterfactual outcome, so the potential-outcomes framework makes counterfactual dependence something a statistical model can describe (Holland 1986).

The account has two well-known difficulties. First, "the closest possible world" is a philosophical notion that has no obvious empirical meaning. Second, it handles redundant causes badly: if two sufficient causes were both present, removing either one leaves the effect in place, so neither counts as a cause. The potential-outcomes framework avoids the first difficulty by defining the counterfactual in terms of a specific action (see below), and mostly sidesteps the second by asking about the effect of one well-specified action at a time.

2.2.3 Manipulation and intervention

Definition 2.3 (Manipulability account of causation)**.** On the **manipulability account**, A is a cause of B if an intervention that changes A, and that affects nothing else that bears on B, would change B (Woodward 2003).

Here the cause is something that someone could in principle change. Holland summarized this view as "no causation without manipulation" (Holland 1986).

This is the account that the book's definitions (Definition 1.1 and Definition 1.6) take on: a cause is an *action*, and the effect is a contrast between outcomes under different actions. The consequence is that questions about attributes we cannot intervene on (for example, "what is the causal effect of sex or of race?") are not well posed until we specify an intervention.

2.2.4 Probabilistic causation

Definition 2.4 (Probabilistic view of causation)**.** On the **probabilistic view**, A causes B if A raises the probability of B, after holding fixed other relevant factors (Suppes 1970).

This view repairs the regularity view for stochastic outcomes. It also matches how epidemiologists work: a risk factor changes the probability of disease and does not determine it. Its main difficulty is the one every observational study faces. Probability raising can come from confounding, so "after holding fixed other relevant factors" is doing the heavy lifting, and the view does not say which factors those are. The later chapters on exchangeability and bias are in part a detailed answer to that question.

2.2.5 Mechanisms

Definition 2.5 (Mechanistic view of causation)**.** On the **mechanistic view**, A causes B if there is a mechanism, an organized set of entities and activities, that links A to B (Machamer et al. 2000).

Mechanistic evidence, such as a known biological pathway, often strengthens a causal conclusion even when the statistical evidence is limited. The potential-outcomes framework does not require a mechanism to define an effect. It defines the effect of an action on an outcome without describing how the action works.

2.3 What the potential-outcomes framework takes from these accounts

Table 2.1 summarizes how each account relates to this book.

Table 2.1: Philosophical accounts of causation and where they appear in this book.

Account	Core idea	Relation to this book
Regularity	Events of type A are followed by events of type B	Motivates asking whether an association is causal; too strict for stochastic outcomes
Counterfactual	B would not have occurred without A	Basis of potential outcomes (Definition 1.3, Definition 1.5)
Manipulability	A cause is something one could change	Basis of the action-based definitions (Definition 1.1, Definition 1.6)
Probabilistic	Causes raise probabilities	Matches estimands defined on risks and means
Mechanistic	Causes work through mechanisms	Not needed for definitions; useful as supporting evidence

Two assumptions in the book come straight from the counterfactual and manipulability accounts.

Consistency (Definition 1.8) says that the outcome we observe under action a equals the potential outcome $Y(a)$. That statement is only meaningful if "action a" names one thing. Hernán and Taubman (2008) and Hernán (2016) argue this using examples such as obesity and body mass index: different ways of changing the exposure can lead to different outcomes, so a question about "the" effect of the exposure is ambiguous until the intervention is specified. VanderWeele and Hernán (2013) formalize the point as the problem of multiple versions of treatment.

Well-defined interventions are the practical form of the manipulability account. If we cannot say what intervention would set A to a, we cannot say what $Y(a)$ refers to, and consistency has nothing to hold on to.

Example 2.2 (A well-defined and an ill-defined question)**.** "What would a patient's blood pressure be if they took 10 mg of a particular antihypertensive daily for a year?" names a specific intervention, so $Y(a)$ is meaningful.

"What would a patient's blood pressure be if they had a lower body weight?" does not. Diet, exercise, surgery, and illness all lower weight, and they plausibly affect blood pressure differently. We would need to specify a particular intervention to define the causal question.

2.4 Not everyone agrees

The manipulability requirement is contested. Some authors hold that causal questions about attributes such as race or sex are meaningful without a specified intervention. Others reply that such questions can only mean something once they are tied to an intervention on a related, manipulable variable. This book follows the potential-outcomes tradition and requires a specified intervention. That choice is a design decision for the course, and the other accounts above remain legitimate frameworks.

2.5 Further reading

- The original discussion of the regularity and counterfactual formulations is Hume's *Enquiry* (Hume 1748).
- Lewis (1973) is the standard statement of the counterfactual account.
- Woodward (2003) develops the interventionist (manipulability) account in detail.
- Holland (1986) connects the philosophical discussion to statistical causal inference.
- Hernán (2016) gives an epidemiologist's argument for well-defined interventions.

3 Collider bias

3.1 Definition

A **collider** is a variable that is caused by two or more other variables in a causal diagram (directed acyclic graph or DAG) (Catalog of Bias Collaboration 2024b). The term "collider" comes from the visual representation where multiple arrows "collide" into the variable.

The basic collider structure is an **inverted fork (V-structure)**:

```
Exposure → Collider ← Outcome
```

More complex patterns exist, such as **M-structures** where multiple colliders are connected in chains, creating additional opportunities for bias when conditioning on any collider in the structure.

Inappropriately controlling for a collider variable, by study design or statistical analysis, results in **collider bias** (Catalog of Bias Collaboration 2024b).

Controlling for a collider can induce a distorted association between the exposure and outcome when in fact none exists. This bias predominantly occurs in observational studies.

Because collider bias can be induced by sampling (restricting the study to certain values of the collider), selection bias can sometimes be considered to be a form of collider bias (Catalog of Bias Collaboration 2024b).

3.2 Classic example: Sackett's hospitalized patients

A classic example of collider bias was provided by Sackett (1979) and discussed by Catalog of Bias Collaboration (2024b). He analyzed data from 257 hospitalized individuals and detected an association between locomotor disease and respiratory disease (odds ratio 4.06). However, when he repeated the analysis in a sample of 2,783 individuals from the general population, he found no association (odds ratio 1.06). The original analysis of hospitalized individuals was biased because both diseases caused individuals to be hospitalized. By looking only within the stratum of hospitalized individuals, Sackett had observed a distorted association. In this example, locomotor disease and respiratory disease are independent causes of hospitalization—the collider. Controlling for the collider by study design (selection bias) induced a distorted association between the two diseases.

3.3 Modern example: The obesity paradox

A more recent example can be seen in the "obesity paradox"—an apparent protective effect of obesity on mortality in individuals with chronic conditions such as cardiovascular disease (CVD) (Catalog of Bias Collaboration 2024b). In fact, obesity increases mortality rates in the general population. The collider bias occurs when investigators condition on CVD (by design or analysis), resulting in a distorted association. Consequently, in a sample that includes only patients with CVD, obesity falsely appears to protect against mortality, whereas in the wider population, obesity increases the risk of early death. Banack and Kaufman (2014) showed using the third US National Health and Nutrition Examination Survey (NHANES III) that the unbiased mortality risk ratio for the entire cohort was 1.24 [95% CI = 1.11, 1.39] (harmful), but the biased stratum-specific mortality risk ratio in patients with CVD was 0.79 [95% CI = 0.68, 0.91] (protective).

3.4 Prevention

Collider bias can be prevented by carefully applying appropriate inclusion criteria—making sure that the exposure and outcome of interest do not drive inclusion or selective retention in a study (Catalog of Bias Collaboration 2024b). Causal diagrams (directed acyclic graphs or DAGs) can help identify colliders that should be left uncontrolled and confounders that should be controlled.

For more detailed information about collider bias, including causal diagrams and additional examples, see Catalog of Bias Collaboration (2024b).

4 Bias analyses

"Continuation of Annual Screening Mammography and Breast Cancer Mortality in Women Older Than 70 Years" (2020), supplementary materials p16:

> A sensitivity analysis for unmeasured confounding can only conclude that, if there is much unmeasured confounding, there will be much bias in the estimates and, conversely, if there is little unmeasured confounding, there will be little bias in the estimates. But, if the magnitude of unmeasured confounding is unknown (as it often is), then the magnitude of confounding bias is also unknown. That is, sensitivity analysis for unmeasured confounding is, by definition, uninformative. The situation is different when investigators have some information about the unmeasured confounder(s). For example, our analysis did not adjust for cigarette smoking, which may be associated with both breast cancer screening (because smokers are less likely to use preventive services) and breast cancer death. If we knew the prevalence of smoking in screening-defined groups and the association between smoking and breast cancer death, then we could use published methods to correct the confounded effect estimates.
>
> Unfortunately, these methods assume that the residual confounding can be summarized by a single variable that happens to be dichotomous, time-fixed, and unassociated with all measured variables that were adjusted for in the analysis (Lash et al. 2009; Schneeweiss 2006).

See also Fox et al. (2022) for an updated reference.

5 Difference in differences analyses

Many approaches to causal inference assume exchangeability (Definition 1.10) and exploit its consequence (Theorem 1.1):

$$\mathrm{E}[Y(x)|X = x'] = \mathrm{E}[Y(x)|X = x]$$

Difference-in-differences makes a weaker exchangeability assumption:

$$\mathrm{E}[Y_t(0) - Y_{t'}(0)|X = 1] = \mathrm{E}[Y_t(0) - Y_{t'}(0)|X = 0]$$

5.1 Change in Changes

The **Change in Changes (CiC)** model (Athey and Imbens 2006) is a DiD-type method that estimates the **Quantile Treatment Effect on the Treated (QTET)** — that is, treatment effects across the full distribution of outcomes, not just the mean (Callaway 2024).

CiC requires two periods of data (a pre-treatment and a post-treatment period). The data can be either repeated cross-sections or panel data (Callaway 2024).

5.1.1 Assumption

Rather than the standard parallel trends assumption (that average outcomes would have followed parallel paths absent treatment), CiC assumes that the *distribution* of untreated potential outcomes evolves over time in the same way for both treated and control groups (Athey and Imbens 2006).

5.1.2 Covariate adjustment

CiC can condition on covariates by first fitting a linear model for outcomes conditional on group–time indicators and covariates, then residualizing (removing predicted values), and finally applying the CiC estimator to these quasi-residuals (Athey and Imbens 2006; Callaway 2024).

5.1.3 R implementation

The `CiC()` function in the `qte` R package (Callaway 2024) implements this estimator. Key arguments are (note: `formla` and `xformla` are the actual argument names in the package):

Argument	Description
`formla`	`y ~ d` where `y` is the outcome and `d` is a binary treatment indicator
`xformla`	Optional one-sided formula for additional covariates (e.g., `~ age + education`)
`t`	Post-treatment time period
`tmin1`	Pre-treatment time period
`tname`	Name of the column containing the time variable
`data`	Data frame containing all variables
`panel`	`TRUE` if the dataset is panel data
`probs`	Vector of quantile levels at which to estimate the QTET
`iters`	Number of bootstrap iterations for standard errors

The function returns a `QTE` object with QTET estimates and (optionally) bootstrap confidence intervals at each quantile in `probs`.

5.1.4 Example

The following example from Callaway (2024) estimates the QTET for the National Supported Work Demonstration job-training program using the `lalonde.psid.panel` dataset, conditioning on several pre-treatment characteristics:

```
library(qte)
data(lalonde)

c1 <- CiC(
  re ~ treat,
  t = 1978, tmin1 = 1975, tname = "year",
  xformla = ~ age + I(age^2) + education + black + hispanic + married + nodegree,
  data = lalonde.psid.panel, idname = "id",
  se = FALSE, probs = seq(0.05, 0.95, 0.05)
)

summary(c1)
#>
#> Overall ATT:
#>  round(ate, 4)
#>        4629.24
#>
#>
#> QTT:
#>   Tau      QTT
#>  0.05    0.000
#>  0.10    0.000
#>  0.15    0.000
#>  0.20    0.000
```

```
#>  0.25  485.230
#>  0.30  929.884
#>  0.35 1460.360
#>  0.40 2321.110
#>  0.45 3462.560
#>  0.50 4232.310
#>  0.55 5010.340
#>  0.60 6210.670
#>  0.65 7458.110
#>  0.70 7508.974
#>  0.75 8210.837
#>  0.80 8332.693
#>  0.85 8515.980
#>  0.90 8821.998
#>  0.95 8273.740
#> ---
#> Signif. codes: `*' confidence band does not cover 0
```

The resulting QTET estimates suggest that the treatment had no measurable effect on the lower quantiles of the earnings distribution but large positive effects at higher quantiles, with an estimated average treatment effect of roughly $4,600 (Callaway 2024).

6 Further resources

6.1 The Catalog of Bias

The Catalog of Bias[1] (Catalog of Bias Collaboration 2024a) is a collaborative project mapping all the biases that affect health evidence. The catalog provides:

- Clear definitions and explanations of various biases
- Real-world examples from health research
- Directed acyclic graphs (DAGs) illustrating bias mechanisms
- Guidance on study design and analysis to prevent or minimize bias

The catalog is an invaluable resource for researchers, clinicians, policymakers, journalists, and the public who seek to better understand threats to research validity.

[1] https://catalogofbias.org/

7 Summary

In summary, this book has no content whatsoever.

```
1 + 1
#> [1] 2
```

References

Athey, Susan, and Guido Imbens. 2006. "Identification and Inference in Nonlinear Difference-in-Differences Models." *Econometrica* 74 (2): 431–97. https://doi.org/10.1111/j.1468-0262.2006.00668.x.

Banack, Hailey R., and Jay S. Kaufman. 2014. "The Obesity Paradox: Understanding the Effect of Obesity on Mortality Among Individuals with Cardiovascular Disease." *Preventive Medicine* 62: 96–102. https://doi.org/10.1016/j.ypmed.2014.02.001.

Callaway, Brantly. 2024. *CiC: Change in Changes Estimator for Quantile Treatment Effects*. https://bcallaway11.github.io/qte/reference/CiC.html.

Catalog of Bias Collaboration. 2024a. *Catalog of Bias*. https://catalogofbias.org/.

Catalog of Bias Collaboration. 2024b. *Collider Bias*. https://catalogofbias.org/biases/collider-bias/.

"Continuation of Annual Screening Mammography and Breast Cancer Mortality in Women Older Than 70 Years." 2020. *Annals of Internal Medicine* 172 (6): 381–89. https://doi.org/10.7326/M18-1199.

Fox, Matthew P., Richard F. MacLehose, and Timothy L. Lash. 2022. *Applying Quantitative Bias Analysis to Epidemiologic Data*. Second edition. Statistics for Biology and Health. Springer. https://doi.org/10.1007/978-3-030-82673-4.

Hernán, M. A., and S. L. Taubman. 2008. "Does Obesity Shorten Life? The Importance of Well-Defined Interventions to Answer Causal Questions." *International Journal of Obesity* 32 (S3): S8–14. https://doi.org/10.1038/ijo.2008.82.

Hernán, MA, and J Robins. 2020. "Causal Inference: What If. Boca Raton: Chapman & Hill/Crc.(2020)." *Publisher's Note Springer Nature Remains Neutral with Regard to Jurisdictional Claims in Published Maps and Institutional Affiliations*. https://miguelhernan.org/whatifbook.

Hernán, Miguel A. 2016. "Does Water Kill? A Call for Less Casual Causal Inferences." *Annals of Epidemiology* 26 (10): 674–80. https://doi.org/10.1016/j.annepidem.2016.08.016.

Holland, Paul W. 1986. "Statistics and Causal Inference." *Journal of the American Statistical Association* 81 (396): 945–60. https://doi.org/10.1080/01621459.1986.10478354.

Hume, David. 1748. *An Enquiry Concerning Human Understanding*. London.

Judea Pearl, Nicholas P. Jewell, Madelyn Glymour. 2016. *Causal Inference in Statistics: A*

Primer. 1st ed. Wiley.

Lash, Timothy L., Matthew P. Fox, and Aliza K. Fink. 2009. *Applying Quantitative Bias Analysis to Epidemiologic Data*. 1st ed. Statistics for Biology and Health. Springer. https://doi.org/10.1007/978-0-387-87959-8.

Lewis, David. 1973. "Causation." *The Journal of Philosophy* 70 (17): 556–67. https://doi.org/10.2307/2025310.

Machamer, Peter, Lindley Darden, and Carl F. Craver. 2000. "Thinking about Mechanisms." *Philosophy of Science* 67 (1): 1–25. https://doi.org/10.1086/392759.

Mackie, J. L. 1965. "Causes and Conditions." *American Philosophical Quarterly* 2 (4): 245–64.

Sackett, D. L. 1979. "Bias in Analytic Research." *Journal of Chronic Diseases* 32 (1-2): 51–63. https://doi.org/10.1016/0021-9681(79)90012-2.

Schneeweiss, Sebastian. 2006. "Sensitivity Analysis and External Adjustment for Unmeasured Confounders in Epidemiologic Database Studies of Therapeutics." *Pharmacoepidemiology and Drug Safety* 15 (5): 291–303. https://doi.org/https://doi.org/10.1002/pds.1200.

Suppes, Patrick. 1970. *A Probabilistic Theory of Causality*. North-Holland.

VanderWeele, Tyler J., and Miguel A. Hernán. 2013. "Causal Inference Under Multiple Versions of Treatment." *Journal of Causal Inference* 1 (1): 1–20. https://doi.org/10.1515/jci-2012-0002.

Woodward, James. 2003. *Making Things Happen: A Theory of Causal Explanation*. Oxford University Press.